CLIENT-SERVER ARCHITECTURE

Recap

- Public, private, and hybrid cloud?
- SaaS, PaaS and IaaS?

Subjects

- Design considerations
- Architectural styles
- System architectures
- Example

Design Considerations

Design Considerations

- Support sharing of resources
  - The network is the computer
- Distribution transparency
- Openness and accessibility
- Scalability

Design Considerations: Distributed Transparency

Transparency	Description
Access	Hide differences in data representation and how an object is accessed
Location	Hide where an object is located
Relocation	Hide that an object may be moved to another location while *in use*
Migration	Hide that an object may move to another location
Replication	Hide that an object is replicated
Concurrency	Hide that an object may be shared by several independent users
Failure	Hide the failure and recovery of an object

Design Considerations: Distributed Transparency

- Aiming at *full distribution* transparency may be too much
  - There are communication latencies that cannot be hidden
  - Completely hiding failures of networks and nodes is (theoretically and practically) impossible
  - You cannot distinguish a slow computer from a failing one
- Full transparency will *cost performance*, exposing distribution of the system
  - Keeping replicas exactly up-to-date with master takes time
  - immediately flushing write operations to disk for fault tolerance

Design Considerations: Distributed Transparency

- *Exposing distribution may be good*
- Making use of location-based services (finding your nearby friends)
- When dealing with users in different time zones
- When it makes it easier for a user to understand what's going on
  - E.g., when a server does not respond for a long time, report it as failing

- Distribution transparency is a nice goal, but achieving it is a different story.

Design Considerations: Openness

Interact with services form other open systems, irrespective of the underlying environment:

- Systems should conform to well-defined interfaces
- Systems should easily interoperate
- Systems should support portability of applications
- Systems should be easily extensible

Design Considerations: Failures

- No single point of failure
- Data replication
- Automated actions on failure
- Logging

Design Considerations: Scalability

At least three components:

- Number of users and/or processes (size scalability)
- Maximum distance between nodes (geographical scalability)
- Number of administrative domains (administrative scalability)

- Most systems account only, to a certain extent, for size scalability.
- Often a solution: Multiple powerful servers operating independently in parallel. Today, the challenge still lies in geographical and administrative scalability

Design Considerations: Scalability

Root causes for scalability problems with centralized solution:

- The computational capacity, limited by the CPUs/GPUs.
- The storage capacity, including the transfer rate between CPUs and disks.
- The network between the user and the centralized service.

Design Considerations: Geographical Scalability

- Can not simply go from LAN to WAN.
  - Many distributed systems assume synchronous client-server interactions:
    - Client sends request and waits or answer. Latency may easily prohibit this scheme.
- WAN links are often inherently unreliable:
    - Simply moving streaming video from LAN to WAN is bound to fail.
- Lack of multipoint communication
    - So that a simple search broadcast cannot be deployed. Solution is to develop separate naming and directory services (having their own scalability problems)

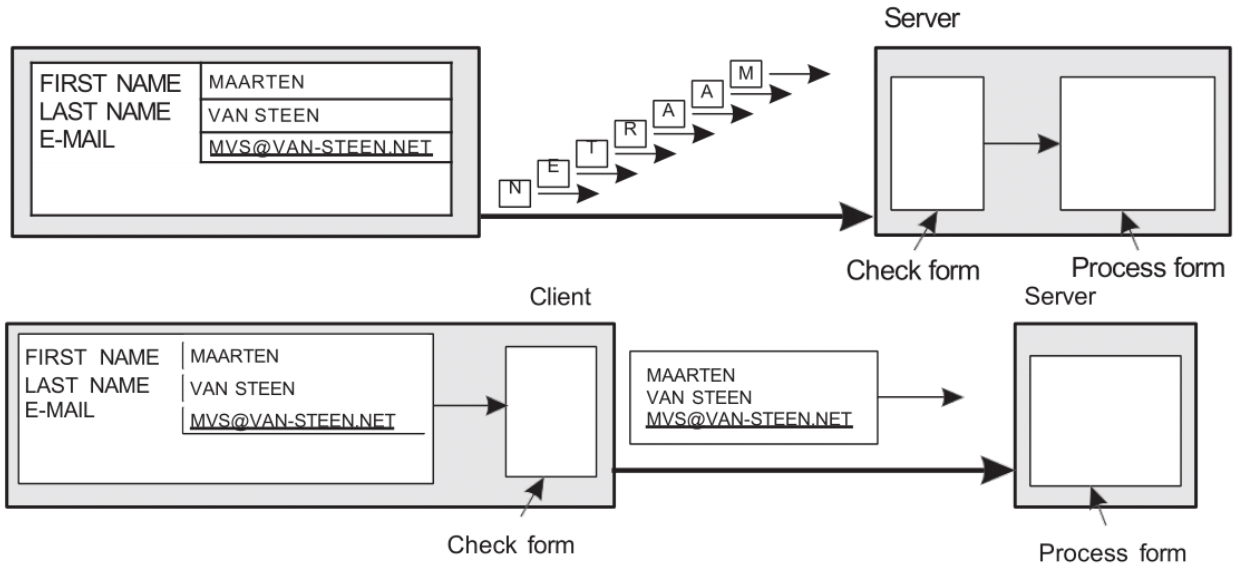
Design Considerations: Administrative Scalability

- For example, conflicting policies concerning usage (and thus payment), management, and security.
    - Typical issue in hybrid clouds.

Design Considerations: Techniques for scaling

- Loose coupling of the components
- Stateless design
- Database choice and design

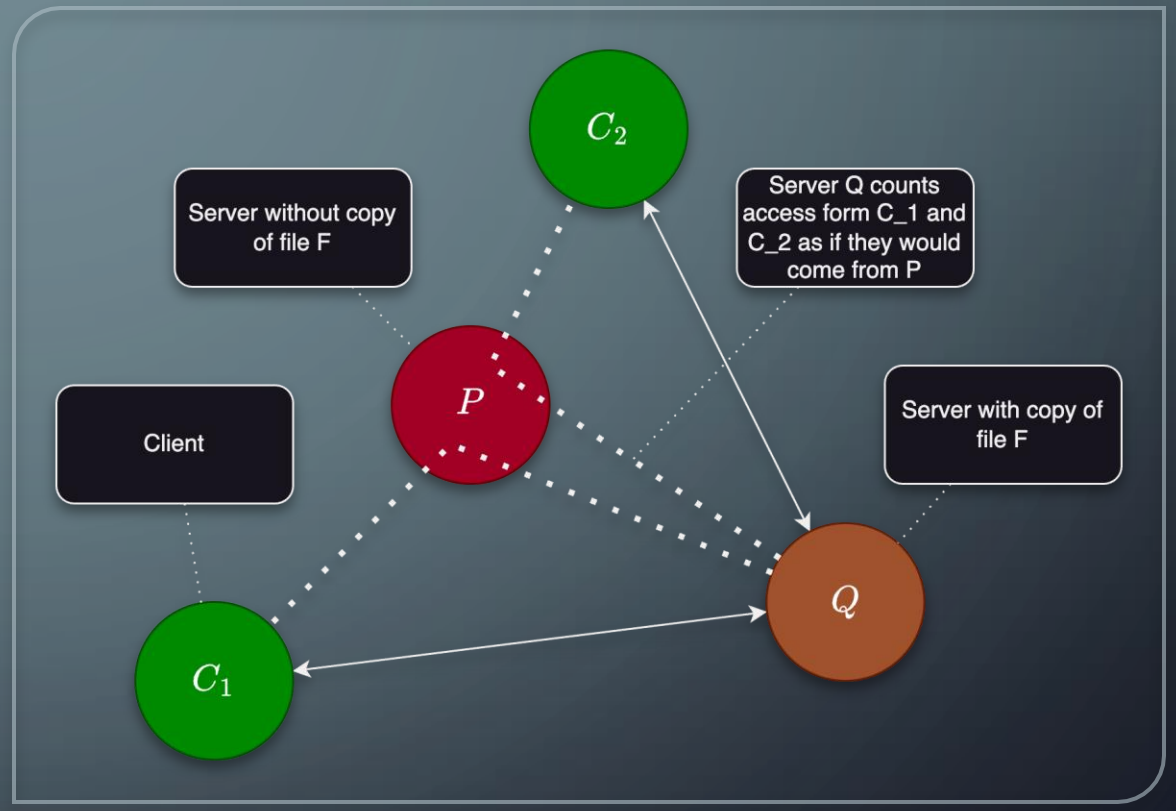
Design Considerations: Techniques for scaling



- Partition data and computations across multiple machines
- Move computations to clients (Java applets)
- Decentralized naming services (DNS)
- Decentralized information systems (WWW)

Design Considerations: Techniques For Scaling

- Replication and caching: Make copies of data available at different machines
  - Replicated file servers and databases
  - Mirrored web sites
  - Web caches (in browsers and proxies)
  - File caching (at server and client)



Architecture

Architecture

- Software architecture
  - Logical organization and interaction of software components
- System architecture
  - Instantiation of a software architecture on real machines

Architectural Styles

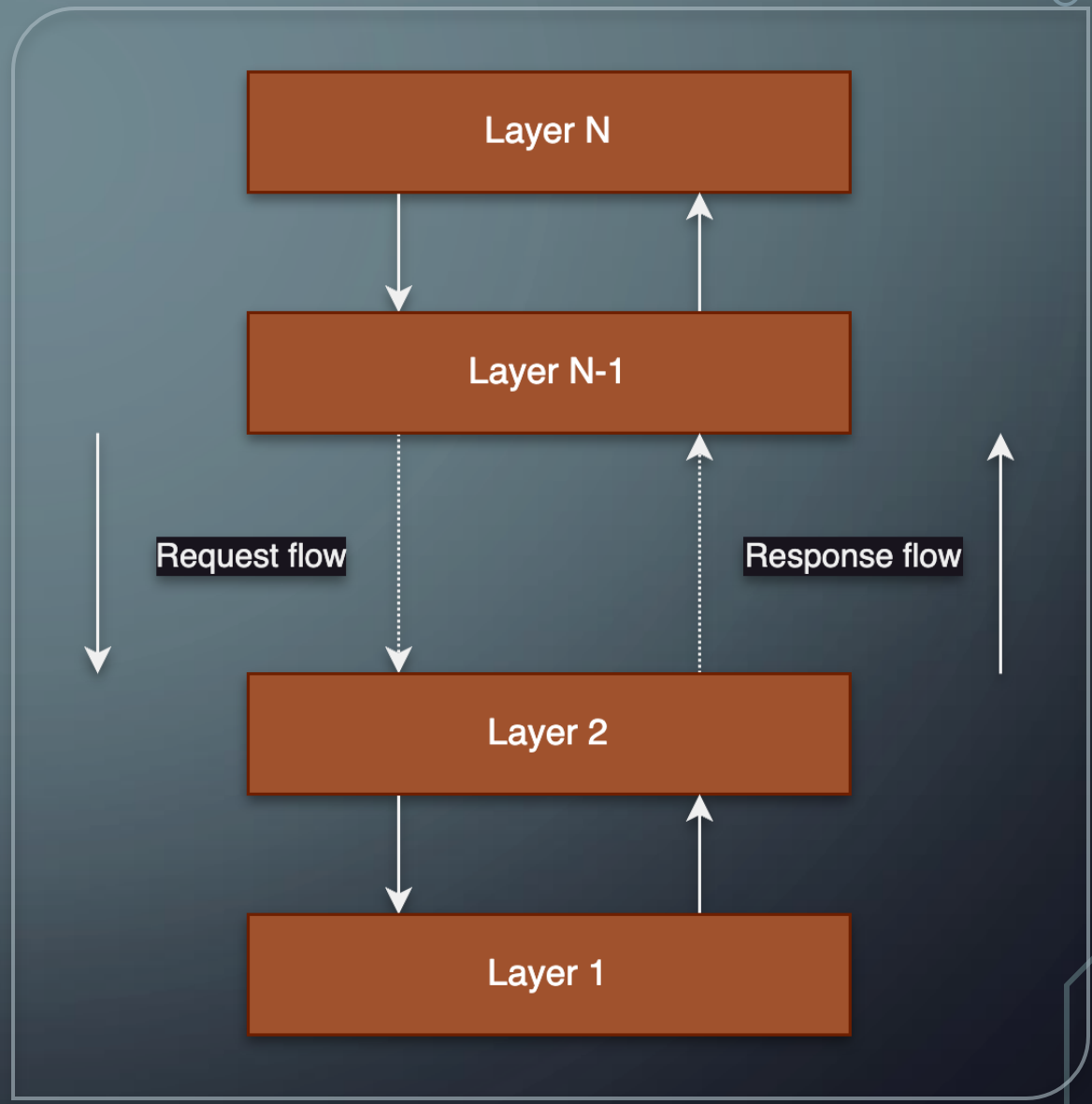
- The notion of an architectural style
  - Formulated in terms of (replaceable) components, their connections and the data exchanged between them
  - A component is a modular unit with well-defined requirements and provided interfaces, replaceable within its environment
  - A connector is a mechanism mediating communication, coordination or cooperation among components

Architectural Styles

- Important architectural styles for distributed systems
  - Layered architectures
  - Object-based architectures
  - Data-centered architectures
  - Event-based architectures

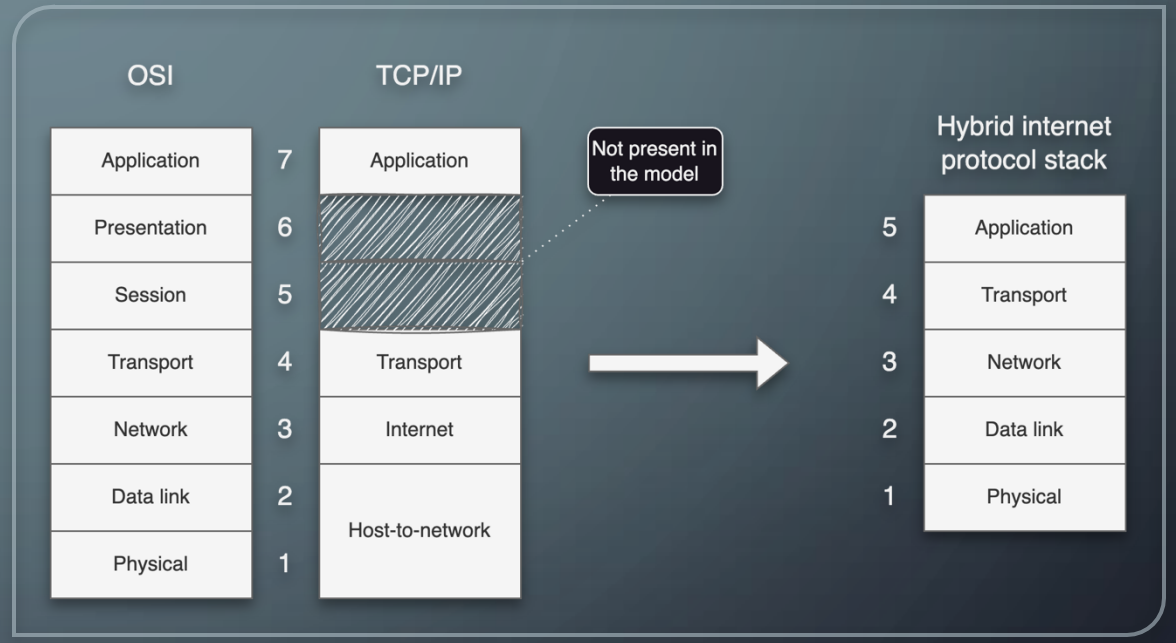
Layered Architectures

- Component at layer L_i is allowed to call component at the underlying layer L_{i-1}, but not the other way around
- Control generally flows layer to layer
  - Request go down
  - Results flow upward
- Widely adopted in network
- Example: OSI model



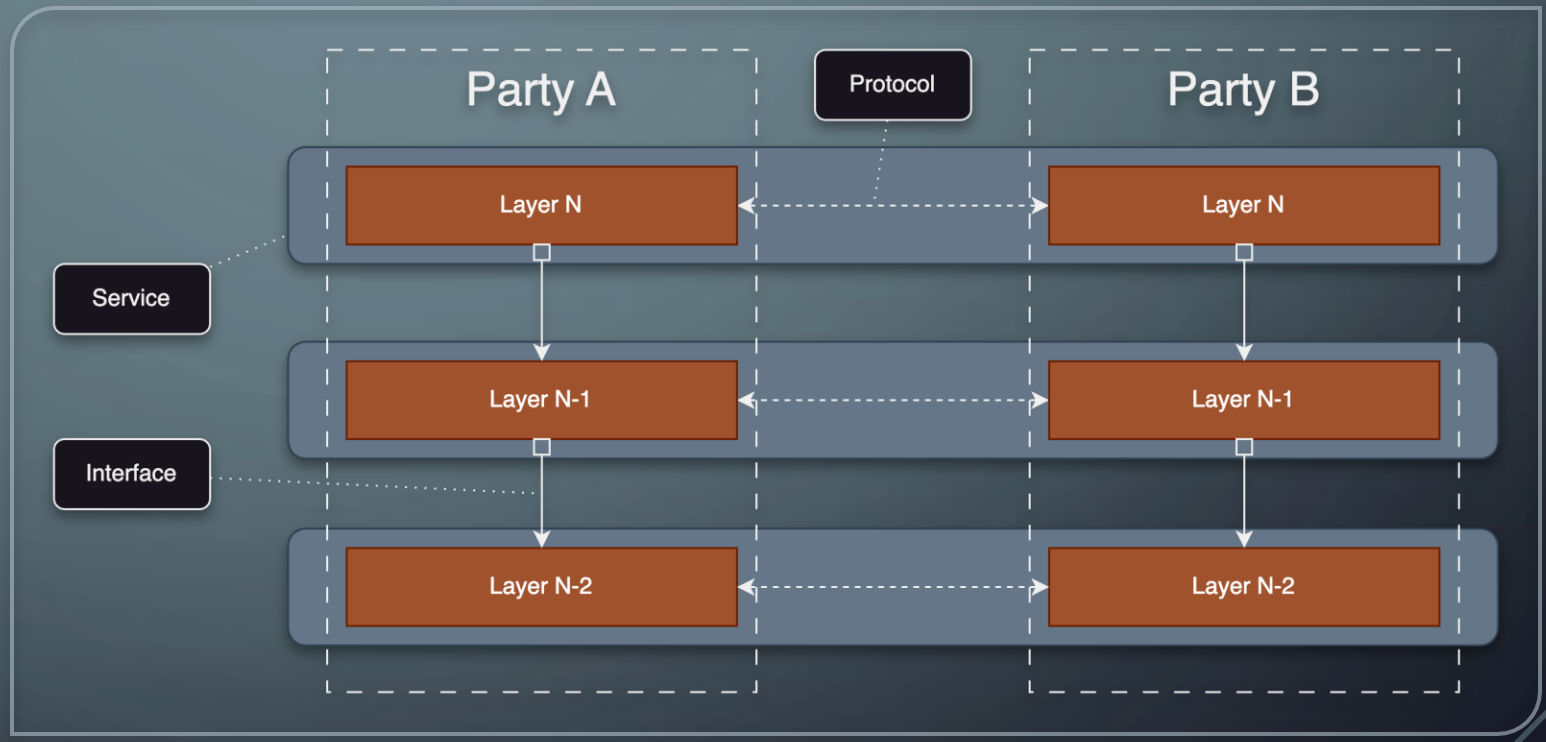
Layered Architectures

- Layered networking architectures
- Layering allows mastering the complexity
  - Explicit structure allows identification and relationship of complex system's pieces
  - Modularization eases maintenance and updating of system
    - Change of implementation of a layer's service transparent to the rest of the system



Layered Architectures

Protocol, service, interface

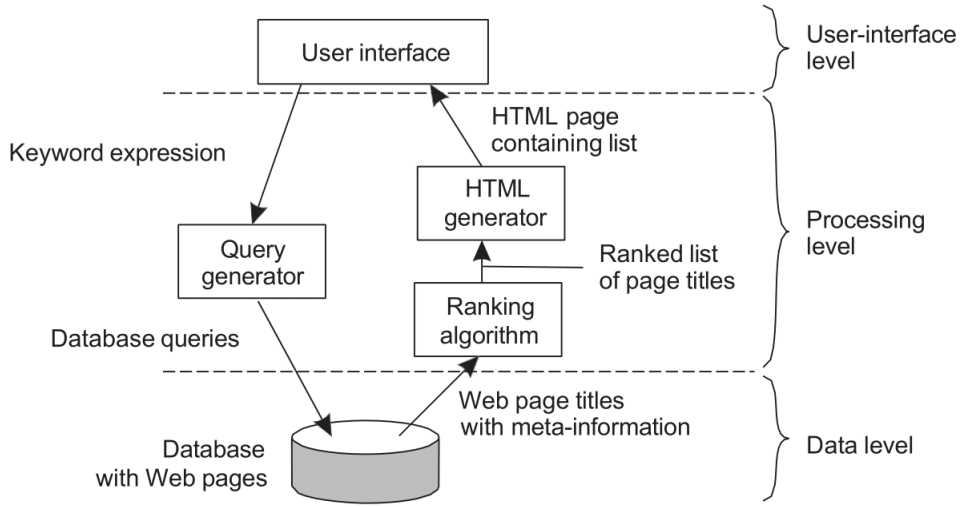


Application Layering

- Application-interface layer contains units for interfacing to users or external applications
- Processing layer contains the functions of an application, i.e., without specific data
- Data layer contains the data that a client wants to manipulate through the application components

APPLICATION LAYERING

EXAMPLE: A SIMPLE SEARCH ENGINE



Application Layering

User-interface Level

- Typically implemented by the client
- Consists of programs that allow end users to interact with applications
- Great variation in functionality provided by the user interfaces

Processing Level

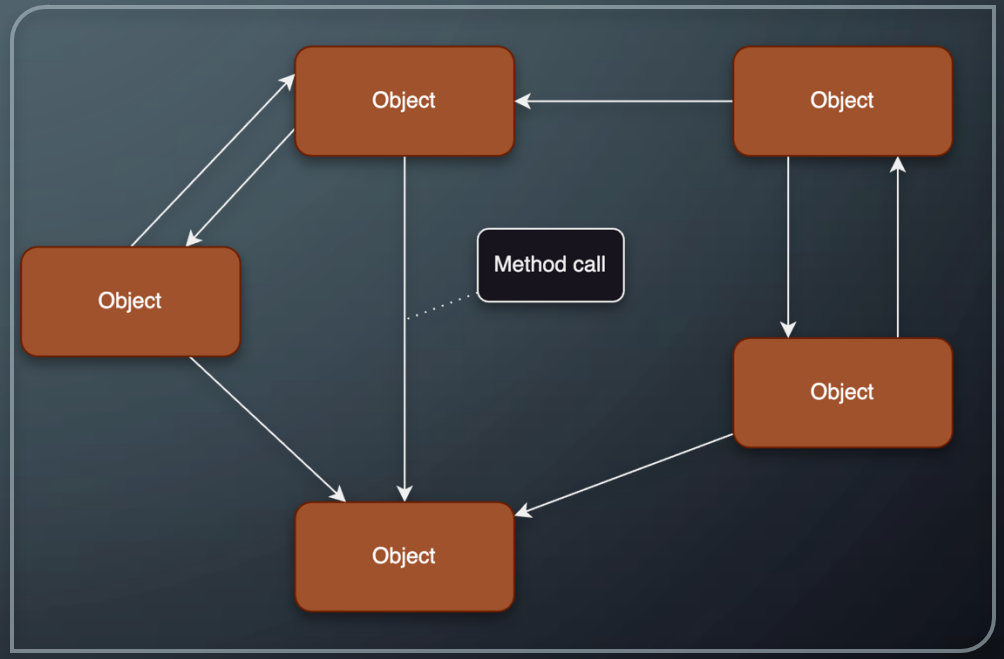
- Contains core functionality of an application

Data Level

- Contains programs that maintain the data on which the application operate
- Persistency
- Consistency

Object-based Architectures

- Objects correspond to components
- Components are connected via a (remote) procedure call mechanism



RESTful Architectures

- View a distributed system as a collection of resources, individually managed by components. Resources may be added, removed, retrieved, and modified by (remote) applications.
  - Resources are identified through a single naming scheme
  - All services offer the same interface
  - Messages sent to or from a service are fully self-described
  - After executing an operation at a service, that component forgets everything about the caller

Basic Operations

Operation	Description
PUT	Create a new resource
GET	Retrieve the state of a resource in some representation
DELETE	Delete a resource
POST	Modify a resource by transferring a new state

Data-Centered Architectures

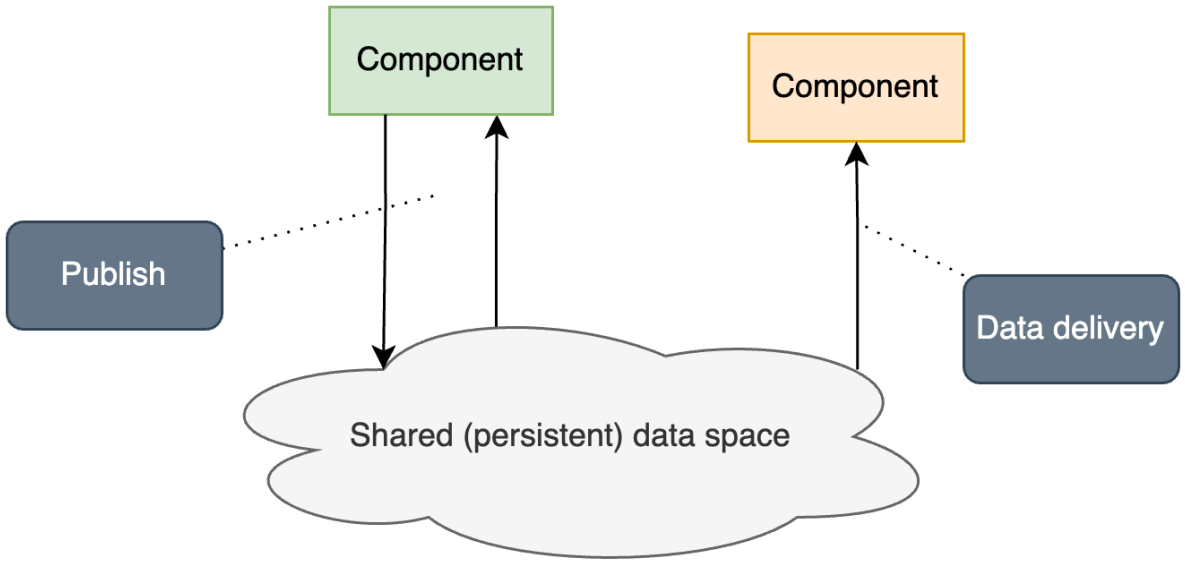
- Processes communicate through a common (passive or active) repository
- Examples
  - Distributed file systems
  - Web-based data services

Temporal and Referential Coupling

	Temporally coupled	Temporally decoupled
Referentially coupled	Direct	Mailbox
Referentially decoupled	Event-based	Shared data space

Event-based Architectures

- Processes communicate through propagation of events
- Events can optionally carry data
- **Publish/subscribe systems**
  - Processes publish events
  - Only processes having subscribes to particular events will receive them
- Allows loose coupling of processing
  - Processes need not explicitly refer to each other (referential decoupling)

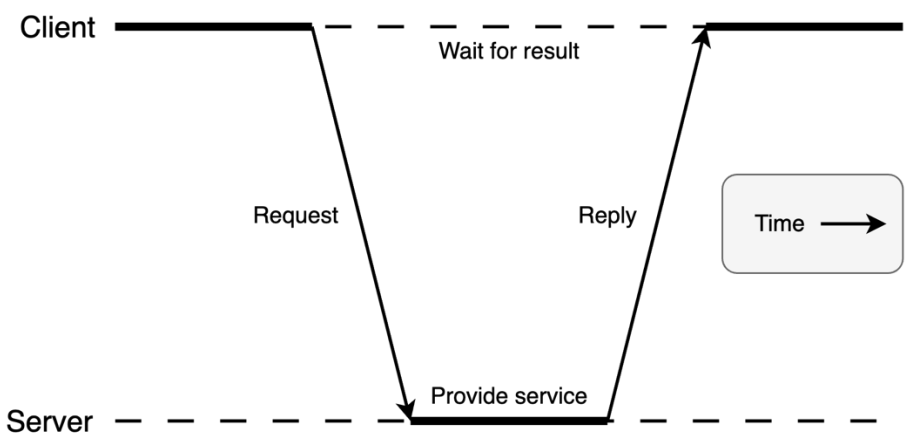


Shared Data Spaces

- Event-based architecture combined with data-centered architecture
- Processes are also decoupled in time (they need not both be active when communication takes place)
- Data can be accessed also using a description instead of explicit reference

System Architectures

- Centralized architectures
- Decentralized architectures (such as Peer-to-Peer)
- Hybrid architectures



Centralized Architectures

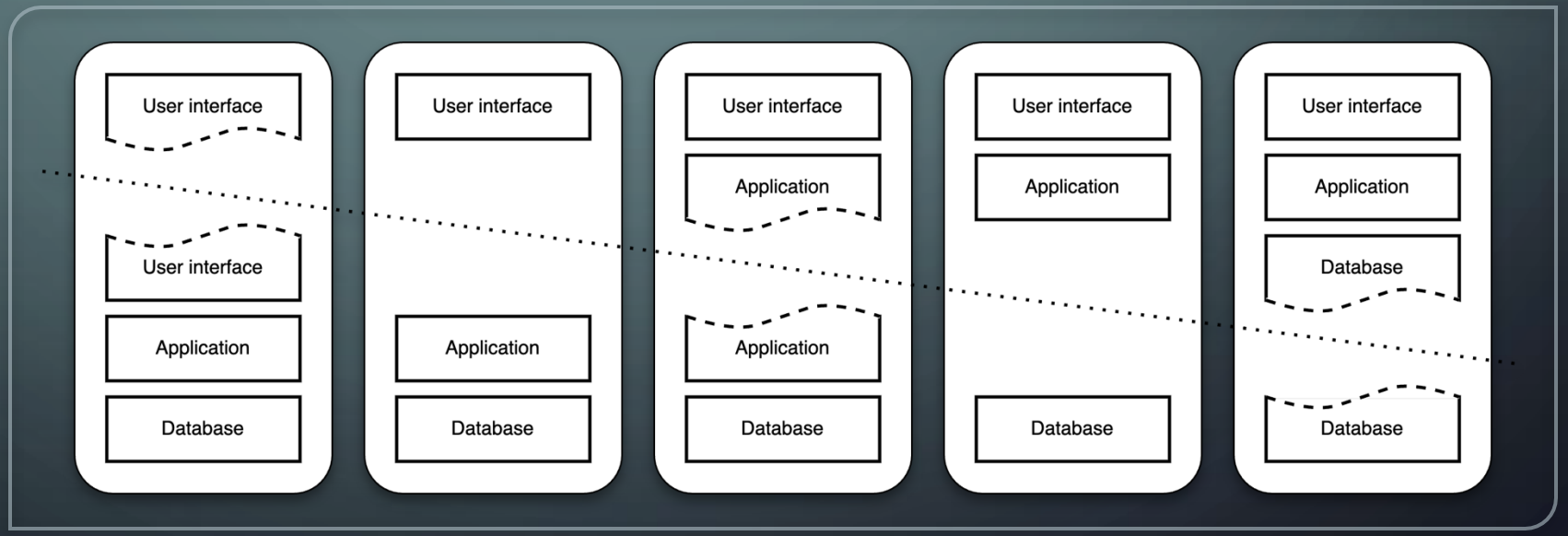
- Client-server model
  - Server
  - Client
- Request-reply behaviour

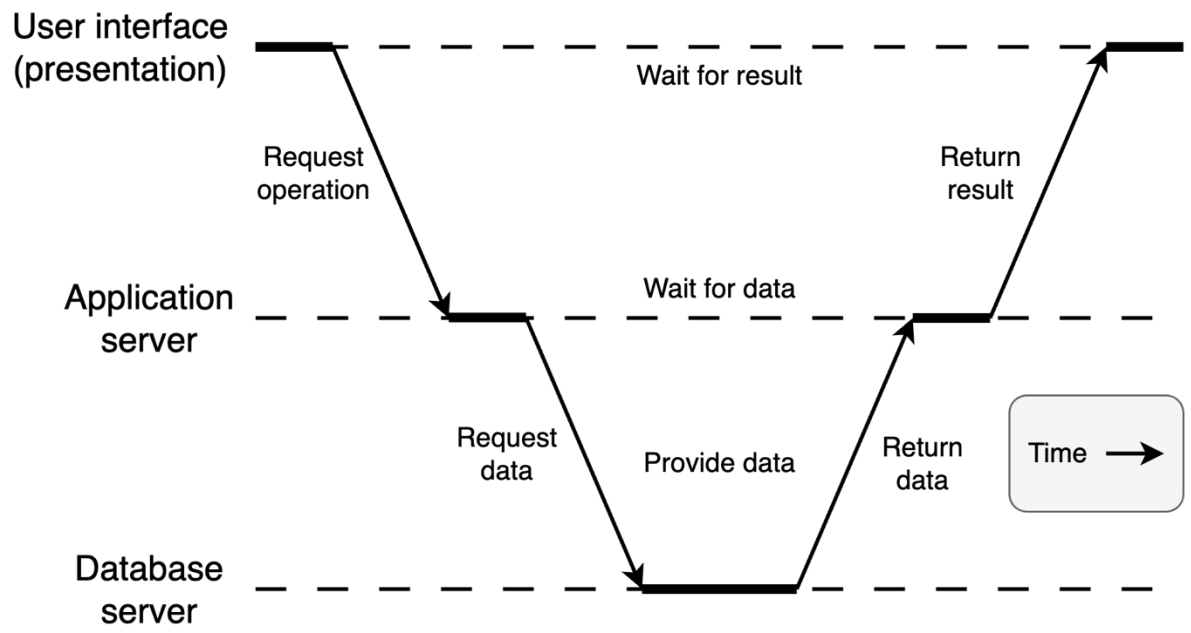
Call Semantics And Transmission Failures

- Ideally: Exactly-once
- Zero-or-more (“maybe”): Service may or may not have been called
- At-least-once: Keep requesting service until valid response arrives at client
- At-most-once: No reply may mean that no execution took place
- Idempotent vs non-idempotent operations
  - Idempotent (repeatable) operation can be repeated multiple times without harm

Multi-tiered Architectures

- Simplest organization is to have only two types of machines
  - Client: Machine containing only the programs implementing (part of) the user-interface level
  - Server: Machine containing the rest (programs implementing the processing/application and data level)



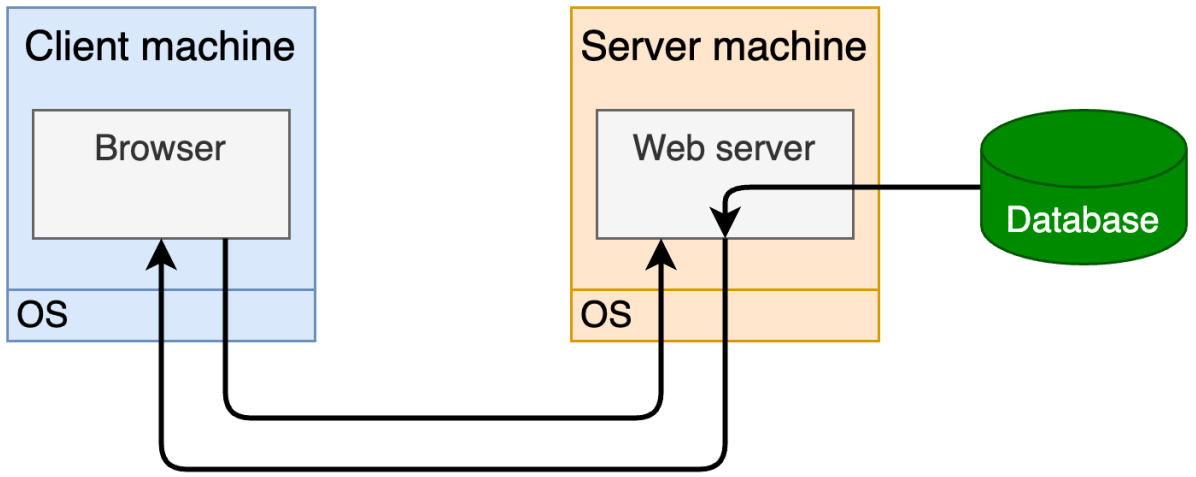


Multi-tiered Architectures

- Three-tiered architecture
  - Single server is replaced by multiple servers running on distributed machines
  - Server sometimes acts as the client
- Vertical distribution
  - Logically different components are placed on different machines

Web: Original Simple Client-server Architecture

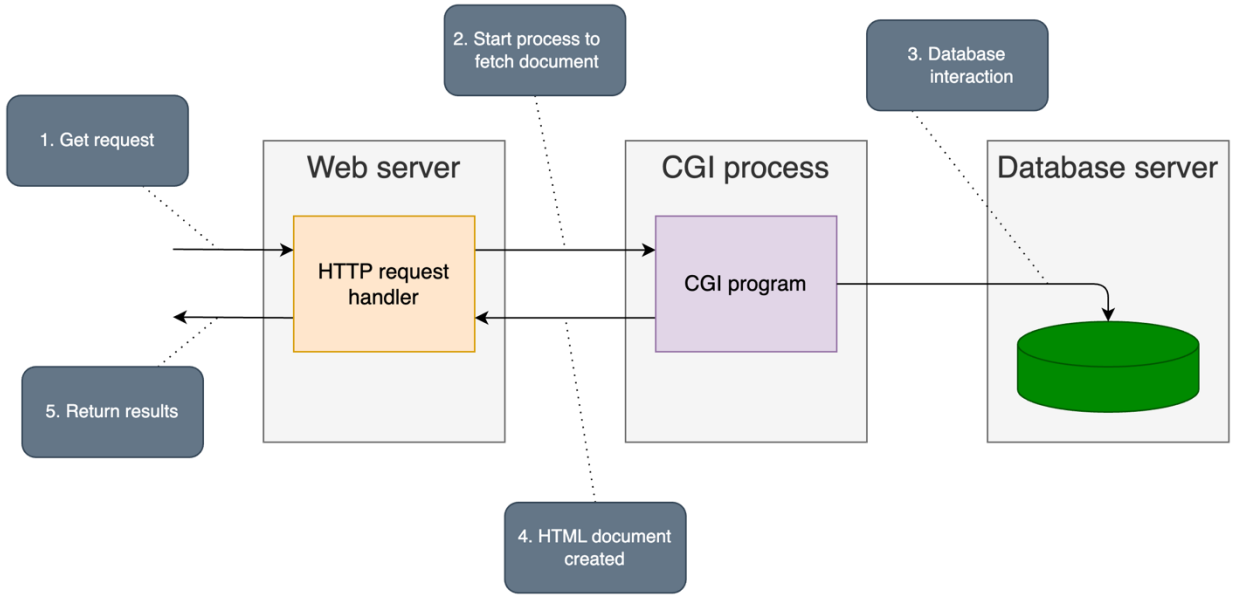
- Server
  - Maintains collection of documents
  - Accepts requests for fetching document and transfers it to client
  - Can also accept requests for storing new documents
- Client interacts with servers through a special application (browsers)



Web: Original Simple Client-server Architecture

Typical operation sequence

1. Browser requests a document referred by means of URL
2. Web server fetches document from local file
3. Web server transmits document to browser
4. Browser displays document



Web: Multi-tiered Architectures (1)

- CGI: Common Gateway Interface
  - Standard way for web server executing a program taking user data as input

Web: Multi-tiered Architectures (2)

- **Server-side script** (JavaScript/NodeJS, PHP, Python, Go)
  - Executed by server when document has been fetched locally
  - Result of executing script (not the script itself) is sent along with the rest of the document to the client
- **Client-side script** (JavaScript)
  - For dynamic generation of web pages at the client
  - Script interpreted and executed by JS engine typically built into the web browser
- **Applets** (e.g., Java applets, ActiveX controls)
  - Pre-compiled stand-alone applications sent to client and executed in browser's address space
- **Servlets** (e.g., JSP, ASP)
  - Pre-compiled program executed in server's address space
- **Helper applications**
  - Independent applications assisting browser in displaying documents
- **Plug-in**
  - Small program dynamically loaded from local file system browser for handling specific document type
- **Web proxy**
  - Originally, support to handle application-level protocols
  - Now, a cache shared by multiple browsers for enhanced performance